Lab Activity: Center of Mass, Torque, and Static Equilibrium

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November 19, 2025

1 Introduction

In this lab, we are investigating *static equilibrium*. A system experiences static equilibrium if the following two conditions are met: (1) The net external force is zero, and (2) the net external torque is zero. In the form of equations, conditions (1) and (2) are

$$\sum_i \vec{F}_i = 0 \quad (1)$$

$$\sum_i \vec{\tau}_i = 0 \quad (2)$$

Consider 1. The pivot S provides enough normal force to support the weight of m_1, m_2, m_3, and the ruler with mass m_r. Thus, the net force is zero and condition (1) is met. To find the net torque, note that $m_1 = 50$ grams, $m_2 = 75$ grams, and $m_r = 150$ grams. We would like to find an m_3 that achieves balance, but how do we calculate the torque due to m_r?

2 The Center of Mass

A *weighted average* for a set of data points x_i and *weights* w_i is calculated as follows:

$$1 = \sum_i w_i \quad (3)$$

$$\bar{x} = \sum_i w_i x_i \quad (4)$$

That is, we ensure the weights sum to 1, and then multiply each data point by a weight, to give it more or less weight in the average. If all the weights are equal, then the weighted average reverts to the arithmetic mean, or regular average. Suppose we want to compute the weighted average location of the ruler, weighted by mass:

$$M = \sum_i m_i \quad (5)$$

$$\vec{r}_{\mathrm{CM}} = M^{-1} \sum_i m_i \vec{r}_i \quad (6)$$

This vector is known as the *center of mass*. Show that, if we chop the ruler into 100 equal pieces, $\vec{r}_{\mathrm{CM}}$ is located approximately where the ruler reads 50 cm.

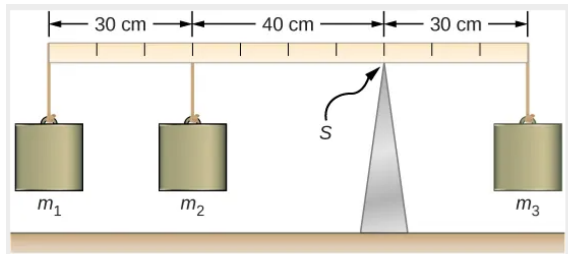

Figure 1: A torque balance.

3 The Net Torque

Now that we know where the center of mass is located, calculate the torque due to m_r by letting the $\vec{r}$ in $\vec{r} \times \vec{F}$ be $\vec{r}_{\mathrm{CM}}$:

Solve for m_3 such that the net torque is zero:

4 Lab Measurement

Recreate the situation in Fig. 1 using the rulers, pivot weights, and masses provided. Record below the locations of the masses, and calculate the net torque. Does the system balance?